

Water percolation through the root-soil interface



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ABSTRACT

Plant roots exude a significant fraction of the carbon assimilated via photosynthesis into the soil. The mucilaginous fraction of root exudates affects the hydraulic properties of the soil near the roots, the so called rhizosphere, in a remarkable and dynamic way. After drying, mucilage becomes hydrophobic and limits the rewetting of the rhizosphere. Here, we aim to find a quantitative relation between rhizosphere rewetting, particle size, soil matric potential and mucilage concentration. We used a pore-network model in which mucilage was randomly distributed in a cubic lattice. The general idea was that the mucilage concentration per solid soil surface increases the contact angle between the liquid and solid phases consequently limiting the rewetting of pores covered with dry mucilage. We used the Young–Laplace equation to calculate the mucilage concentration at which pores are not wettable for varying particle sizes and matric potentials. Then, we simulated the percolation of water across a cubic lattice. Our simulations predicted that above a critical mucilage concentration water could not flow through the porous medium. The critical mucilage concentration decreased with increasing particle size and decreasing matric potential. The model was compared with experiments of capillary rise in soils of different particle size and mucilage concentration. The experiments confirmed the percolation behaviour of the rhizosphere rewetting. Mucilage turned hydrophobic at concentrations above 0.1 mg/cm². The critical mucilage concentration at matric potential of −2.5 hPa was ca. 1% [g/g] for fine sand and 0.1% [g/g] for coarse sand. Our conceptual model is a first step towards a better understanding of the water dynamics in the rhizosphere during rewetting and it can be used to predict in what soil textures rhizosphere water repellency becomes a critical issue for root water uptake.

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1. Introduction

The rhizosphere is the layer of soil next to plant roots that is actively modified by root growth and exudation [17,21]. This layer of soil has an extent of millimetres up to a maximum of a few centimetres and it has a profound impact on soil hydrology. In fact, all the water that flows through the soil-plant-atmosphere continuum, which corresponds to 40% of the terrestrial precipitation [5], flows across the rhizosphere. Sposito [36] defined the flow of water through the rhizosphere as *productive green water* and suggested that rhizosphere processes are an essential element for sustainable and efficient use of soil-water resources.

The rhizosphere is where plants and soil meet and interact, and its physical, chemical and biological properties differ from those of the adjacent bulk soil [21]. In this study we focus on the physical properties of the rhizosphere and how they affect the water flow through the root-soil interface. Our starting-point are recent observations

of water repellency in the rhizosphere [13,29]. Carminati et al. [13] found that the rewetting rate of the rhizosphere after a drying/wetting cycle was markedly slower than that of the bulk soil. They found that the rhizosphere of lupines subject to a drying cycle and then irrigated by capillary rise remained drier than the bulk soil for 1–2 days. Moradi et al. [29] showed that the slow rewetting of the rhizosphere was caused by the high contact angle of dry rhizosphere. Carminati [10] observed that rhizosphere hydrophobicity was more marked when the soil was let dry until the plants started to show wilting symptoms, which happened at volumetric water contents below 0.05. However, water repellency in the rhizosphere was also visible when the samples were kept relatively wet (at volumetric water contents above 0.10–0.15). These observations suggest that rhizosphere water repellency is more evident in dry soils, but it also occurs in a broader range of water contents.

Carminati [11] proposed that the water repellency of the rhizosphere resulted from mucilage exuded by roots. Mucilage is primarily exuded at the root tips and consists mainly of polysaccharides and a few percentages of phospholipids [18,28,30]. Guinel and McCully [18] reported that mucilage of maize (*Zea mays*) had a remarkable ability to swell and adsorb water and found that at saturation the

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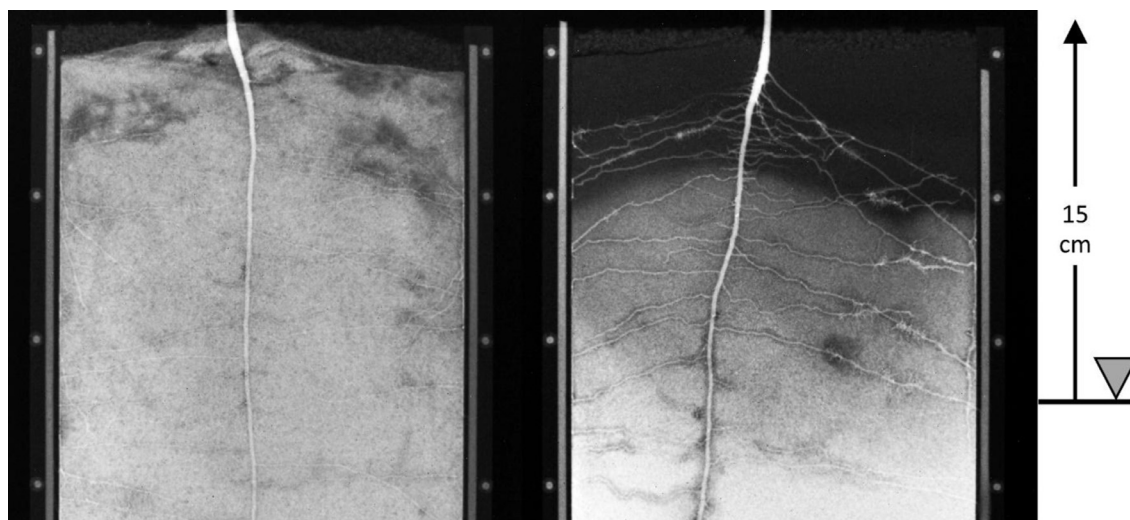


Fig. 1. Neutron radiographs of water distribution around the roots of lupines after irrigation. Left: lupine in a sandy soil; right: lupine in a coarse quartz sand. The samples were rewetted by capillary rise, with the water table at a depth of 15 cm from the soil surface. The grey values are proportional to the water content (dark=dry, bright=wet). Dry zones are visible around the roots, in particular in the coarse quartz sand (right).

weight of wet mucilage is 1000 times higher than its dry weight. Read et al. [31] measured similar mucilage water contents for fully hydrated maize mucilage. Impressed by the ability of mucilage to absorb water, McCully and Boyer [28] tested whether mucilage can hold large volumes of water also when the water potential decreases. They found that most of the water stored in fully hydrated mucilage was actually lost at water potentials less negative than -10 kPa. Nevertheless, Carminati [11] argued that the remaining mucilage water content is sufficient to increase the soil moisture of a few percentages. Young [38] came to a similar conclusion after measuring a higher water content in the rhizosphere of wheat compared to the adjacent bulk soil. Recently, Ahmed et al. [1] and Kroener et al. [26] showed that mucilage exuded by chia seeds increases the water content of a sandy soil at any water potential if the system is in hydraulic equilibrium. Similar observations were made by Deng et al. [16], who investigated the effect of mucilage exuded from the seed coating of *Capsella bursa-pastoris* L. Medik. on the water retention curve of a sandy clay loam. They found an increase in the saturated water retention due to the presence of seeds and/or mucilage. The increase in water content was visible for water potentials less negative than -10 kPa.

Although these studies suggest a positive effect of mucilage on increasing the water content in the rhizosphere, other studies showed an apparently opposite behaviour. Read and Gregory [32] observed that mucilage surface tension was smaller than the one of water. Read et al. [30] explained this reduction in surface tension as the effect of phospholipids. The reduction of surface tension is likely to decrease the capillary forces and to reduce the soil water content at negative water potential, as shown for an analogue of mucilage phospholipids (lecithin) mixed with a sandy loam [30]. The importance of the surface tension of mucilage is currently poorly understood, but it is likely to play an important role on rhizosphere hydrology, in particular in shaping the initial penetration of mucilage through the soil pores.

Upon drying, the phospholipids present in mucilage are expected to induce some degree of water repellency. In fact, Czarnes et al. [14] observed that soils mixed with polygalacturonic acid (one of the main components of mucilage) became water repellent. Hallett et al. [19] investigated the impact of four plant species on the hydraulic properties of rhizosphere soil and observed different magnitudes of water repellency among the samples. Furthermore, they suggested the idea that the rewetting rate of mucilage was reduced by mucilage swelling

and consequent pore clogging. Moradi et al. [29] measured contact angles higher than 90° in the rhizosphere of lupine roots. Carminati et al. [13] used neutron radiography, to image the water content distribution around the roots of lupines in a sandy soil. They found that during drying the rhizosphere was wetter than the bulk soil. On the contrary, after irrigation the rhizosphere remained dry and it rewetted in a few days. Carminati [11] suggested that these two apparently contrasting behaviours are not in contradiction, but they are rather the result of the dynamic and time-dependent effect of mucilage on the hydraulic properties of porous media. At equilibrium conditions mucilage is capable of holding large volumes of water and it increases the soil water content at any water potentials [26]. After drying mucilage turns hydrophobic and delays the rhizosphere rewetting for a period of hours up to a few days [11], resulting in time-dependent and hysteretic hydraulic properties of the rhizosphere [26].

In the present study we investigate the water dynamics in the rhizosphere during the initial stage of the rewetting phase after the soil was let dry. Our objective was to quantitatively relate the rewetting kinetics of the rhizosphere to mucilage concentration, soil particle size and soil water potential.

To better motivate our study we refer to Fig. 1, which shows the water content distribution around the roots of lupines a few minutes after rewetting subsequent to a severe drying. The grey values are proportional to the water content (dark=dry, bright=wet). The images were obtained using neutron radiography. The details of the experiments are described in Carminati [10]. The samples were 30 cm high and were irrigated by capillary rise, with the water table being placed at 15 cm below the soil surface. The lupine plant on the left side was grown in a sandy soil composed of coarse sand (6.59%), medium sand (49.9%), fine sand (31.5%), coarse silt (3.35%), medium silt (1.8%), fine silt (1.81%) and clay (5.05%). The lupine on the right side was grown in quartz sand with a particle diameter of 0.2 to 0.63 mm. The radiographs show that in both samples the water content in the direct vicinity of roots was reduced, but the reduction was much more marked and evident in the coarser quartz sand (sample on the right side). Note that the reduction in water content near the roots was caused by the altered contact angle of the rhizosphere and could not be trivially explained as the effect of water depletion due to root uptake. Indeed, the soil water content in the two samples was high enough to allow a fast redistribution of water in the soil. The water that roots took up from the rhizosphere was very quickly replaced by water flowing from the bulk soil adjacent to the rhizosphere.

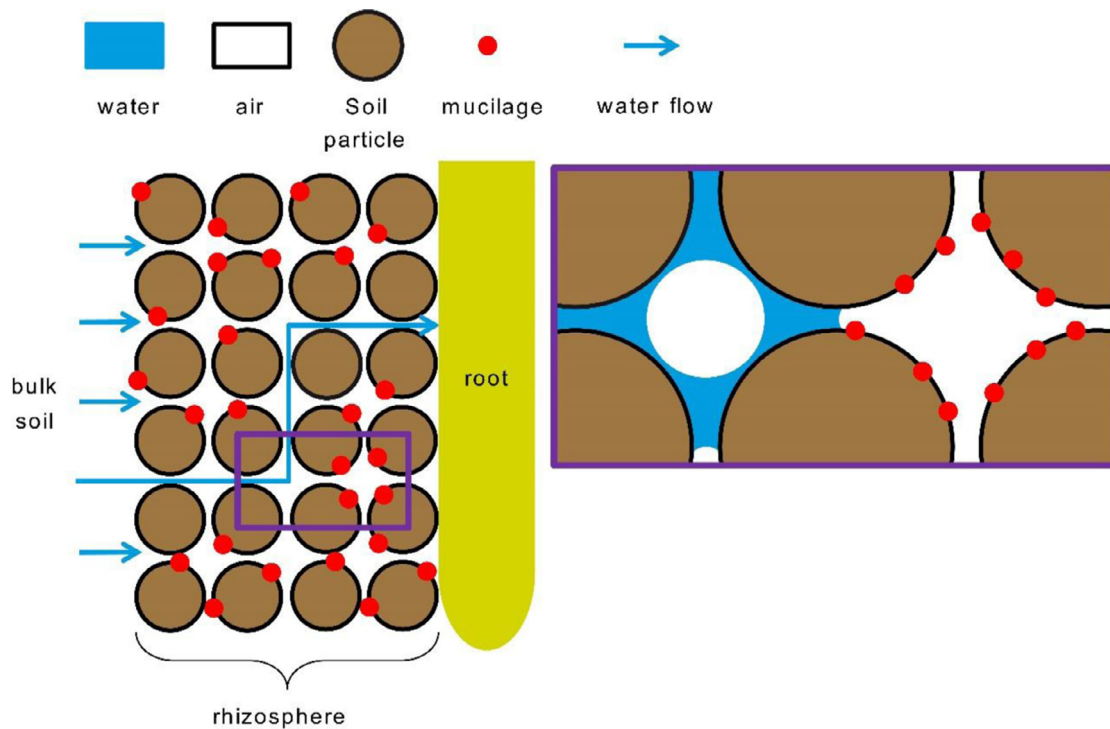


Fig. 2. Illustration of the effect of mucilage on water flow in the rhizosphere. After drying mucilage becomes hydrophobic and impedes the water flow in a fraction of the pores.

Moderate water depletion around the roots could only be expected in drier soil conditions and not at such high water contents.

Our objective was to develop a conceptual model to explain the differences between the rhizosphere of the two samples shown in Fig. 1. We aimed to find the relation between rhizosphere rewetting, soil particle size, mucilage exudation and soil matric potential. We intended to answer the following questions. At what mucilage concentrations does a soil with a given particle size become water repellent? What matric potential is needed to rewet a rhizosphere with a given mucilage concentration and particle size?

To answer these questions we: (1) performed capillary rise experiments with soils of varying particle size and mucilage concentration; (2) developed a simple model of water percolation across the rhizosphere.

The concept behind the model is that the distribution of mucilage at the pore-scale is heterogeneous. This microscopic heterogeneity of mucilage deposition on the pore surface results in a distribution of contact angles in the porous medium. The mucilage concentration on the pore surface affects the contact angle between water and the solid surface and determines the wettability of a pore at a given water potential. If the fraction of wettable pores is high enough to form a connected network spanning throughout the sample water could flow across the porous medium; which is a typical percolation problem.

Percolation theory has been used to predict macroscopic transport properties of porous media by simulating relevant processes at the microscopic scale [6–8,22]. Steenhuis et al. [35] used the percolation approach to explain the experimental observation that a few hydrophobic grains can make a soil water repellent. We followed a similar percolation approach to predict how the microscopic distribution of mucilage at the pore-scale affects the macroscopic rewetting of the rhizosphere.

To test our conceptual model, we used neutron radiography to monitor the water flow through soils and quartz sands of varying particle size mixed with different quantities of mucilage. Additionally,

we measured the contact angle of glass slides covered with varying quantities of mucilage.

2. Conceptual model

Mucilage is exuded from the root tips and it diffuses into the soil until it binds to the surface of soil particles. It is expected that during the first drying phase mucilage keeps the rhizosphere wet. As roots take up water and the soil dries out, mucilage dehydrates, turns hydrophobic and causes high water repellency in the rhizosphere. At this point, even when the bulk soil is rewetted the rhizosphere remains dry. Afterwards, mucilage slowly rehydrates, the contact angle relaxes and the rhizosphere rewets. The rewetting of the rhizosphere may vary from a period of some hours up to a few days [11]. In this study we focus on the initial phase of the rewetting process and we do not consider mucilage swelling and the rewetting of the rhizosphere. This is justified by the different time-scales of the two processes: the initial water flow through the rhizosphere has a time scale of seconds to a few minutes, the rehydration of mucilage has a time scale of hours.

An illustration of the model is shown in Fig. 2. Dry mucilage is randomly distributed on the surface of the soil particles. The quantity of mucilage on the pore surface affects the contact angle between water and the solid particle. We assume that the contact angle α depends on the mucilage concentration on the pore surface C_S [g/cm²] – i.e. $C_S = \alpha(C_S)$. We expect that the contact angle increases with increasing mucilage concentration per soil surface C_S . It follows that the wettability of a pore depends on both, the specific pore surface and the quantity of mucilage in the pore. We also assumed that mucilage distribution and soil particle position is constant over time.

The wettability of a pore is estimated according to the Young–Laplace equation. A pore is rewetted when the following condition is satisfied:

$$h > \frac{-2\gamma \cos(\alpha)}{r\rho_w g} \quad (1)$$

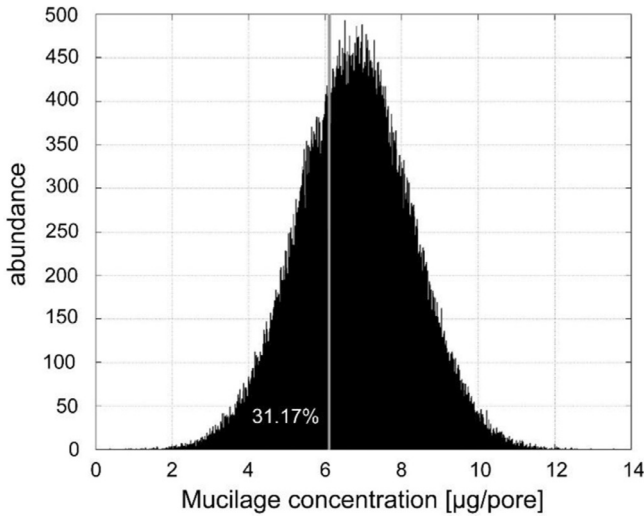


Fig. 3. Example of mucilage distribution in the pore-network model. According to the percolation theory, at least 31.17% of the pores must be conductive to enable the water flow from one side of the system to the opposite side. Eq. (1) allows to calculate if a pore of given radius and mucilage concentration C_5 is wettable at a given matric potential h .

where h is the matric potential (or capillary pressure) expressed in centimetre heads [cm] (note that in unsaturated conditions h is negative), γ is the surface tension of water [J m^{-2}], α is the contact angle, ρ_w is the density of water [g cm^{-3}], r is the pore radius [cm] and g is gravity [m s^{-2}]. By using the Young–Laplace equation to estimate the wettability of a pore, we assume that pores are cylindrical and we ignore corner and film flows in more realistic porous media. Recent and more advanced pore-network models can be found in Blunt [9], Valvatne and Blunt [37], and Joekar-Niasar and Hassanizadeh [23]. Here we used a very simple representation of the porous medium because of the lack of experimental information on the microscopic distribution of mucilage in the pores. Since this information is missing, we decided to start with a simple model.

Similarly, we simplified the pore-network as a cubic lattice, with each cube representing a single pore. Assuming that the soil was a cubic packing of spheres, we obtained that the radius of the embedded pore was 0.73 times the radius of the soil particles.

Then we randomly distributed the mucilage in the cubic packing. A random, normally distributed quantity of mucilage per pore was generated. The distribution was shifted towards the positive

direction by summing to all values the largest absolute value of the distribution. Then the values were scaled to obtain the desired quantity of mucilage in the packing. A distribution of mucilage per pore is shown in Fig. 3.

Based on the mucilage distribution and the particle radius, we calculated the surface mucilage concentration C_5 in each pore. Given the relation between contact angle α and C_5 , we used Eq. (1) to predict if a pore of a given radius r , at a matric potential h is rewettable or not. According to Stauffer [34], the percolation threshold for a cubic lattice is 31.17%. In other words, at least 31.17% of the sites must be conductive to allow the water flow. The line in Fig. 3 shows the distribution of mucilage at the percolation threshold: the pores on the left side of the line satisfy Eq. (1) and are wettable, while the pores on the right side are not wettable.

After the distribution of mucilage, each pore was checked for possible invasion. The process of invasion was started from one face (starting face) of the cubic lattice to its opposite side. A pore was invaded when Eq. (1) was satisfied and the pore was connected to a wetted pore (for each pore we considered the 6 neighbours). An illustration of the process close to the percolation threshold is shown in Fig. 4.

In Fig. 4 we showed the invasion of water from the left side for varying mucilage concentration, C [g/g], defined as the gram of mucilage per gram of dry soil. At a critical mucilage concentration, C_{th} , the infiltration front became irregular and formed fingers, as often observed in hydrophobic soils [4]. Slightly above and below C_{th} the final water content differed largely.

Objective of the model was to predict C_{th} as a function of particle size and soil matric potential. Note that as the soils used in the experiments had a narrow pore size distribution, in the model we assumed a uniform pore size. Of course, our model is a clear simplification of water flow in porous media. Beside ignoring film flow and snap off mechanisms, we also simplified the pore network as a cubic lattice, which has a coordination number equal to 6, while the coordination number of real porous media is variable [37]. We intentionally kept the model as simple as possible, so that we could investigate the effect of a few parameters, soil particle size and mucilage concentration on the water repellency in the rhizosphere.

3. Material and methods

To reproduce an analogue of the rhizosphere, we mixed soils of different particle size with a varying quantity of mucilage, obtaining a range of mucilage concentrations C (dry mass of mucilage per dry mass of soil). We extracted mucilage from the seeds of chia

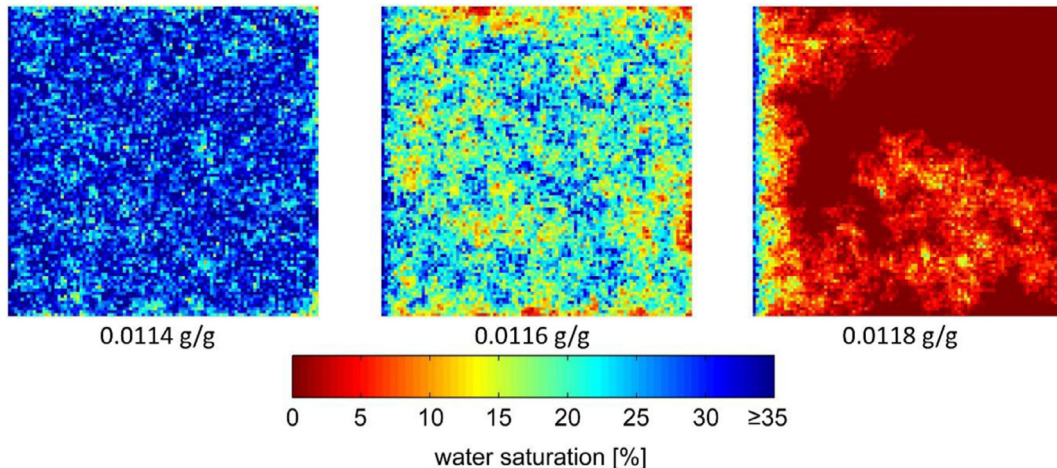


Fig. 4. Simulation of water percolation through a soil with varying mucilage concentrations. The simulations show the invasion of water from the left to the right in a soil with pore radius $r=0.1625$ mm at matric potential $h=-2.5$ cm. Near C_{th} small changes in mucilage concentrations resulted in a large change in water saturation.

Table 1

Soil particle diameters and estimated average surface area A assuming a cubic packing of spheres.

Soil particle diameter [mm]	Surface area A [cm ² /g]
0–0.02	2308
0.063–0.125	245
0.125–0.2	142
0.2–0.355	83
0.355–0.5	54
0.5–0.63	41
0.63–0.1	28

(*Salvia hispanica* L.). We chose mucilage from chia seeds since it can easily be extracted and it has a similar chemical composition to lupine and maize mucilage (mainly xylose, glucose and uronic acids) [27]. Additionally, it also has similar physical properties - i.e. it becomes a gel after immersion in water and it turns hydrophobic after drying [1,26]. Mucilage was extracted according to the procedure described in Ahmed et al. [1] and Kroener et al. [26].

By mixing various amounts of wet mucilage with dry soil, we obtained samples with varying C . The gravimetric water content of the mixture was around 25% to ensure a homogeneous mucilage-soil mixture. Experiments were conducted with a silty soil (particle diameter $d=0-0.02$ mm) and quartz sands of different particle diameters ($d=0.063-0.125$ mm, $d=0.125-0.2$ mm, $d=0.2-0.355$ mm, $d=0.355-0.5$ mm, $d=0.5-0.63$ mm, and $d=0.63-1$ mm).

Assuming that the soil particles had spherical shape and uniform particle size distribution, we estimated that the surface area per gram of soil is $A = \frac{6}{d} \frac{1}{\rho_{mi}}$ [cm²/g], where d is the particle diameter and ρ_{mi} is the mineral density (2.6 g/cm³). The specific surface area A of the different soils is given in Table 1.

The mucilage-soil mixtures were dried for about 24 h at 40 °C. To avoid accumulation of dry mucilage on the surface, the mixture was spread on a large plate with the height of the mixture being smaller than 1 mm. The mucilage-sand composition formed a crust which was then poured through a sieve to separate individual grains.

The experimental set-up consisted of thin rectangular containers of various heights and base of 1.3 cm x 1.3 cm. The containers were filled with three soil layers. The lower part of each sample was filled with untreated soil. The second layer was filled with the different soil-mucilage mixtures. This layer had a thickness of 0.5 cm and represented the rhizosphere. The upper layer consisted of untreated sand. The samples were initially dry. Then the samples were immersed in water until the water table reached a given height below the second layer (the rhizosphere). The tested distances between the water table and the rhizosphere were: 2.5 cm, 6.5 cm and 12 cm (Fig. 5).

The water content in the samples was monitored in real time using neutron radiography. Neutron radiography is an optimal non-invasive method to measure the spatial distribution of water in thin samples with high accuracy [12]. Neutron radiographs were performed at the NEUTRA and ICON imaging stations of the Paul Scherrer Institute, Switzerland [24].

Objective of the neutron radiography experiments were: (1) to investigate whether there was a critical mucilage concentration C_{th} [gram of dry mucilage per gram of dry soil] above which water could not percolate through the rhizosphere layer; (2) to experimentally estimate C_{th} for varying soil-mucilage mixtures and matric potential.

Then we tested if our percolation model was able to reproduce the observed C_{th} . To fit the values of C_{th} , we let vary the relation between the contact angle α and the surface mucilage concentration C_S . Note that α is the microscopic contact angle at the pore-scale.

An initial estimation of the microscopic contact angle as a function of mucilage concentration, $\alpha(C_S)$, was obtained from capillary rise experiments between two glass slides. The glass slides were

prepared with specific amounts of mucilage. Diluted wet mucilage was uniformly spread on the glass slides and it was let dry at 40 °C for 24 h. The slides were then taped together on one corner, while a clip was placed between them in the other corner. In this way a narrow angle formed between the slides. The slides were then placed in dyed water and the capillary rise was monitored. The experiments were conducted several times, with values of C_S ranging from 0 to 5.46×10^{-4} g/cm². Values between these concentrations were chosen with 5.46×10^{-4} g/cm² multiplied by 1/3, 1/9, 1/16, 1/27 and 0.

The capillary rise experiments between the glass plates gave an initial range of values for the relation between α and C_S . Within this range, we determined the curve $\alpha(C_S)$ that gave the best fit between the model and the neutron radiography experiments.

To validate the model, we additionally measured the curve $\alpha(C_S)$ using the sessile drop method (SDM). Slides with specific amounts of dry mucilage were prepared as described above. Then a water droplet was placed with a needle on the slides and the initial contact angle was measured optically at 25 °C (Drop Shape Analyser DSA25S; KRÜSS GmbH). We measured the contact angle for samples with a surface concentration C_S [g/cm²] ranging from 0.00015625 to 0.32 mg/cm². Measurements were repeated at least 10 times for each C_S . These measurements provide the relation between contact angle and surface mucilage concentrations of C_S .

4. Results & discussion

The neutron radiographs showed that at a critical mucilage concentration C_{th} there was a sudden transition in the wettability of the soil-mucilage layer. At mucilage concentrations below C_{th} water could rapidly flow through the layer, while at concentrations above C_{th} the treated layer impeded the water flow for more than one hour. Exemplary radiographs of the experiments are shown in Fig. 5. The images show the water content in the samples 5 minutes after irrigation from the bottom. The water content was derived by referencing the actual radiographs to the radiograph of the dry samples. The detailed image analysis is described elsewhere in details [12,20]. The colours in the images show the water contents. The threshold mucilage concentration C_{th} at the transition between the conductive and the non-conductive state depended on particle size and matric potential. C_{th} decreased with increasing particle size and with decreasing matric potential. For a matric potential of -2.5 cm and particle diameter of 0.125–0.2 mm C_{th} was around 1% [g/g]. For particle diameters of 0.2–0.355 mm at the same matric potential of -2.5 cm C_{th} was 0.75%. At a matric potential of -6.5 cm C_{th} was around 0.375% for 0.125–0.2 mm particle diameter and 0.015% for 0.2–0.355 mm particle diameter.

The simulation of the capillary rise experiments using the pore-network model showed a good agreement with the neutron radiographs. The threshold concentration C_{th} as derived from the neutron radiography experiments and as predicted by the pore-network model are shown in Fig. 6. The effect of particle size and matric potential on C_{th} was well described by the model. Soils with large particle diameters have a small specific surface - i.e. A is proportional to $1/d$. This results in high concentration of mucilage per surface area C_S and high contact angles. Additionally large particle diameters decrease the capillary forces (Eq. (1)), resulting in a lower critical mucilage concentration C_{th} . Similarly, according to Eq. (1) a decrease in matric potential results in a decrease in C_{th} ; in other words, at more negative water potentials, less mucilage is needed to impede the water flow.

The relation between α and C_S is plotted in Fig. 7. The initial estimation of the relation between α and C_S was derived from the capillary rise experiments between two glass plates (region between the two lines). The experiments showed that for 2.02×10^{-5} g/cm² < C_S < 1.82×10^{-4} g/cm² there was no capillary rise between the slides, indicating that $\alpha \geq 90^\circ$. At the other extreme, we found that for $C_S <$

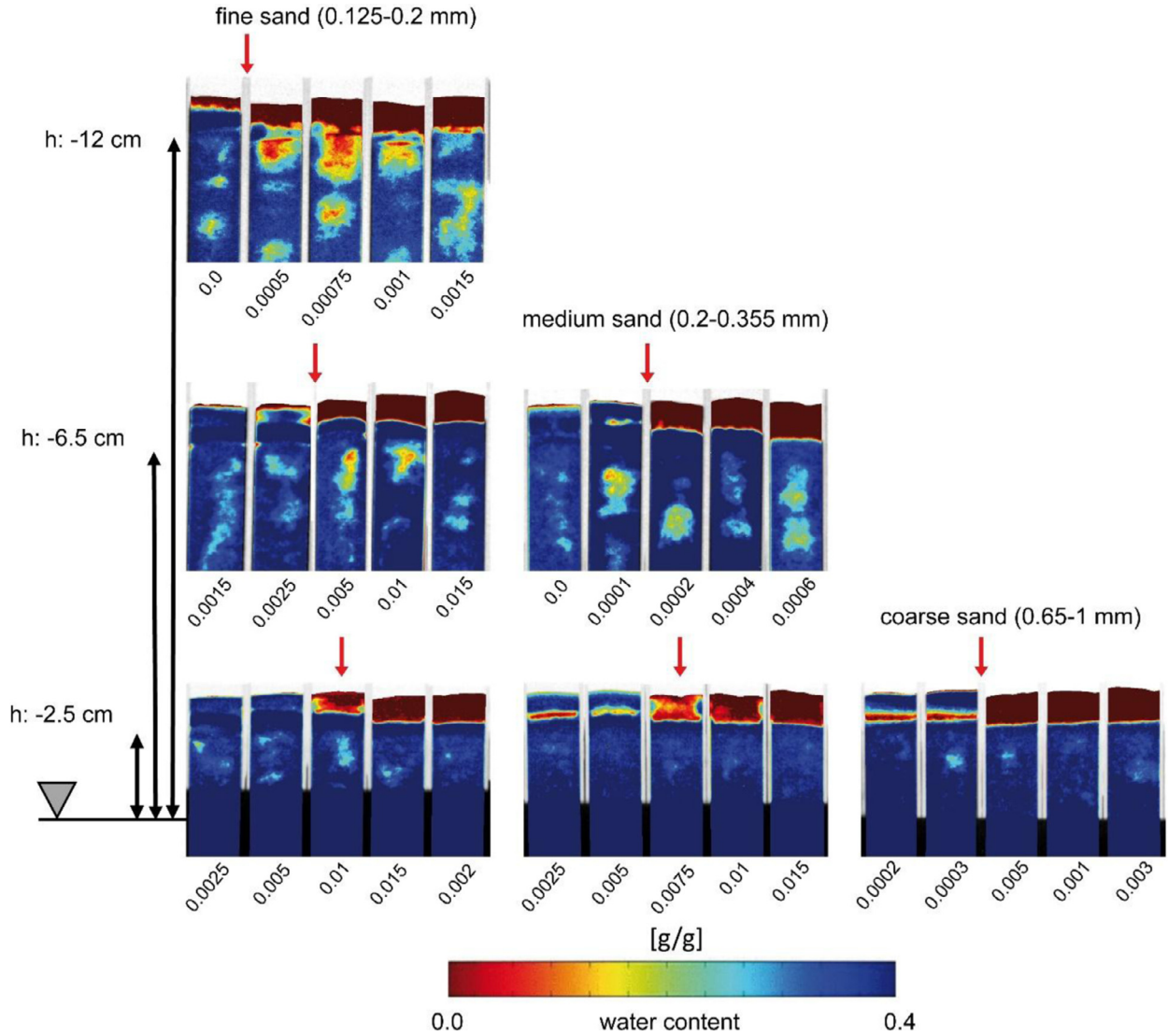


Fig. 5. Neutron radiographs of the capillary rise experiments in fine sand (particle diameter 0.125–0.2 mm), medium sand (0.2–0.355 mm) and coarse sand (0.65–1 mm) at varying matric potentials, $h = -2.5$ cm, $h = -6.5$ cm and $h = -12$ cm. Each sample was prepared with a specific amount of mucilage. The threshold mucilage concentrations C_{th} are indicated with an arrow.

6.75×10^{-6} g/cm² there was no visible influence of mucilage on capillary rise; hence the original contact angle of the used glass plates was unaltered. A minimum contact angle of 30° was chosen for the untreated sandy soil based on the measurements of Moradi et al. [29], who measured the contact angle of the same sands used here – i.e. we took the values of the contact angle far from the root surface where exudates did not alter the soil wettability. These observations gave a range of possible values for the curve $\alpha(C_S)$. The semi-solid line shows the curve $\alpha(C_S)$ that gave the best fit between the pore-network model and the neutron radiography experiments. The relation consists of two polynomial functions:

$$\alpha = -4.05 \cdot 10^{12} C_S^2 + 2.40 \cdot 10^7 C_S + 30$$

$$\text{for } 0 \leq C_S \leq 2.2 \cdot 10^{-6} \text{ g/cm}^2$$

$$\alpha = 1.21 \cdot 10^9 C_S^2 + 1.96 \cdot 10^5 C_S + 62.85$$

$$\text{for } C_S > 2.2 \cdot 10^{-6} \text{ g/cm}^2$$

The measurements with the sessile drop method (crosses) showed a similar trend, with contact angles of 35°–45° at $C_S = 0.00015625$ mg/cm² and contact angles up to 90° and larger for C_S of 0.08 mg/cm². In total 9 of 15 measurements at a concentration

of 0.08 mg/cm² resulted in contact angles $> 90^\circ$, with a maximum of 109°. In the figure we showed only the values $\leq 90^\circ$, because for our model contact angles equal to 90° or bigger than 90° have the same effect.

These results show that the percolation model was able to predict the critical mucilage concentration at which the rhizosphere turned impermeable. The fitted curve $\alpha(C_S)$ felt well in the range of the observations. The effect of particle size and matric potential on C_{th} was well described by the model.

For soils with a particle diameter $d < 0.125$ mm the model overestimated C_{th} by a factor of 2–3. For particle diameters of $d = 0$ –0.02 mm, the experimental C_{th} was 0.094 g/g and the simulated C_{th} was 0.02 g/g. For particle diameters of $d = 0.063$ –0.125 mm, the experimental C_{th} was 0.03 g/g, while the simulated C_{th} was 0.09 g/g. This overestimation may be caused by the swelling of mucilage in the fine pores and the consequent pore clogging, as suggested by Hallett et al. [19]. In fact, assuming that the swelling of mucilage follows a diffusion behaviour, we can expect that the time needed to clog a pore goes like $t = r^2/D$ where r is the pore radius and D is a diffusion coefficient describing the mucilage swelling. In soils with fine particles the time needed to clog the pores would therefore be much

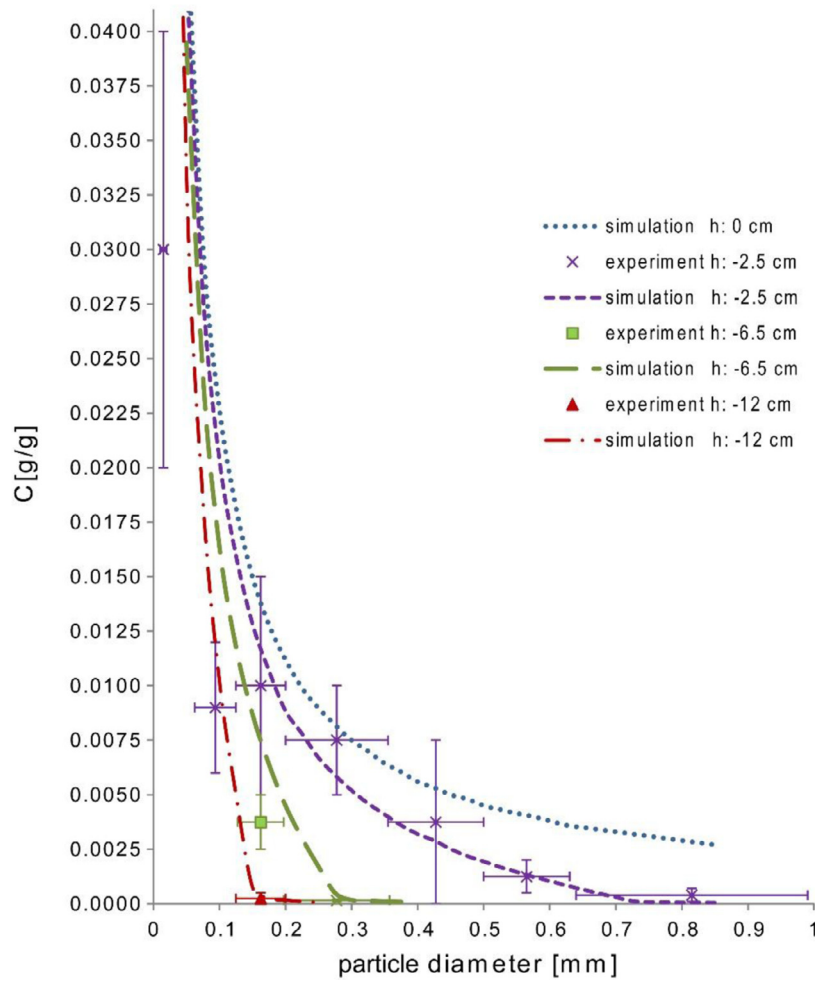


Fig. 6. Mucilage concentration at the percolation threshold (C_{th}) [g of mucilage per g of dry soil] as a function of soil particle diameter for varying matric potentials (h). The lines show the critical mucilage concentrations derived from the pore-network simulations. The points with error bars show the experimental results for different soil particle diameters.

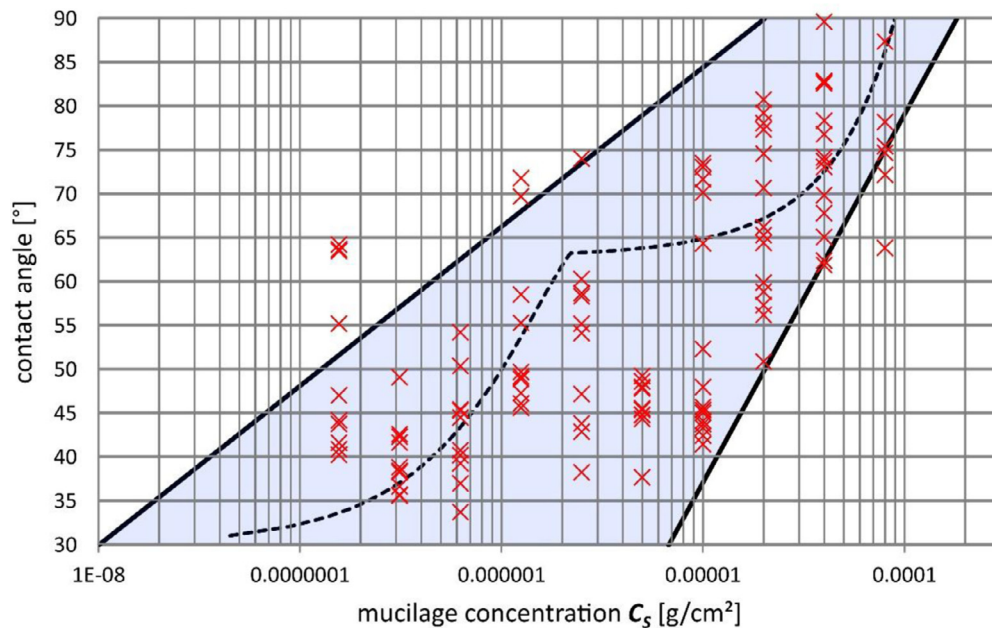


Fig. 7. Measured and fitted relations between mucilage concentration per surface area (C_s) and contact angle (α). Contact angles measured with the sessile drop methods are marked with crosses. The dashed line gave the best fit between observed and simulated C_{th} .

shorter than in coarse textured soils. Based on this argument, we expect that pore clogging due to mucilage swelling occurred during the experiments with the fine textured soils. Due to the joint effect of hydrophobicity and pore clogging, less mucilage was actually needed to impede the flow in the fine textured soils. To describe the effect of pore clogging due to mucilage swelling, the kinetics of the mucilage expansion through the pores should be included in the model. In this paper, we neglected these dynamic effects and focused on the initial rewetting phase.

The model was compared to experiments with a silty soil and quartz sands of comparably narrow grain size distributions. However, natural soils have broader and often multi-modal pore size distributions. In such cases, the spatial distribution of mucilage becomes important. For instance, the possibility that mucilage is uniformly distributed on all pore surfaces or that, more likely, it is preferentially distributed in small or large pores would affect the quantity of mucilage needed to impede the water flow at varying matric potentials. Interestingly, Albalasmeh and Ghezzehei [2] observed a preferential distribution of polygalacturonic acid (an analogue of mucilage) in the contacts between spherical particles that were let dry. Similarly, it could be expected that a higher amount of mucilage is deposited in the small pores during drying. However, our current understanding of pore-scale mechanism of mucilage deposition at varying soil moisture conditions is still limited and it deserves further experimental and modelling studies.

Some of the assumptions of our model were clear simplifications of real porous media. We neglected water flow in pore corners and we used a cubic lattice, which has a coordination number of 6, while more realistic pore-network models could allow variable coordination numbers [37]. Additionally, we focused on the initial phase of the rewetting process and we just checked if water could flow or not through the porous medium, without simulating the time dynamics of the rewetting.

Further improvements of the model should include the implementation of a grain size distribution, for example using a site-bond percolation approach and distributing the different pore radii across the lattice. Secondly, the effects of the relation between pore radius and mucilage quantity should be investigated, either assuming a uniform or a preferential mucilage distribution in pores of a given class. By introducing a more realistic pore geometry, for instance using triangular pores [9], the model could be extended to simulate water and mucilage interactions in the pore corners, where probably mucilage is deposited as the soil dries. Also, tomography based three-dimensional images of the rhizosphere would provide a more precise description of the process. Recently, X-ray CT has been successfully used to reconstruct the pore morphology in the rhizosphere [3,25,33,15]. These advancements in pore-scale imaging offer new avenues to relate the pore-scale physics of the rhizosphere to root water uptake and soil-plant interactions.

5. Conclusions

We developed a simple pore-network model of water percolation through the rhizosphere. The model is capable of reproducing the dual behaviour of the rewetting of soils mixed with mucilage. When the mucilage concentration is below a given threshold C_{th} water can easily cross the rhizosphere. When the mucilage concentration is higher than C_{th} the soil becomes water repellent, impeding the water flow. Near the critical mucilage concentration, the infiltration front becomes irregular and shows fingering.

The simplicity of the model allows highlighting two principle factors controlling the wettability of the root-soil interface: the particle size and the mucilage concentration (gram of mucilage per gram of soil). The radiographs shown at the beginning of this paper (Fig. 1) suggest that the rhizosphere of lupines in sandy soils is near the percolation threshold, where small variations in mucilage

exudation would mean a big change in the wettability of the rhizosphere. It is tempting to think that the amount of mucilage exuded is functional to maintaining the rhizosphere near the percolation threshold, so that the roots can efficiently adjust the hydraulic conductivity of the root-soil interface. To confirm this hypothesis, it would be interesting to study whether plants adapted to different soil textures exude a different amount of mucilage, or, alternatively, a mucilage containing a different amount of hydrophobic compounds. The simplicity of this model allows addressing in a quantitative way such complex ecological questions.

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